

Product Requirements Document — Lexora AI

Version: 0.1 (Draft)
Status: Pre-development
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1. Product Vision

Lexora AI is a professional Legal AI platform for **South African law firms**. It combines matter management, document discovery, email review, legal research, AI-assisted drafting, citation verification, workflows, time tracking, and billing in a single desktop-grade application.

The platform must behave like a **legal work assistant**, not a casual chatbot. Every AI output is traceable, citable, and subject to human review before external use.

1.1 Deployment Philosophy

Priority	Approach
First	Private deployment per law firm — full isolation
Later	Optional hosted multi-tenant SaaS edition

Each private deployment includes:

- Dedicated PostgreSQL database
- Dedicated document storage (MinIO / S3-compatible)
- Dedicated vector database (Qdrant)
- Dedicated full-text search (OpenSearch)
- Dedicated email connectors
- Dedicated audit logs and backups
- Dedicated AI model configuration
- Dedicated domain or subdomain

Architecture must remain **multi-tenant-capable** so a hosted edition can be added without rewrite.

1.2 Target Users

Persona	Primary needs
Partner	Matter oversight, approvals, billing review
Attorney	Drafting, research, matter work, time capture
Candidate Attorney	Supervised drafting, research, task execution
Paralegal	Document prep, discovery, filing support

Persona	Primary needs
Secretary	Calendar, correspondence, document management
Billing Admin	Time approval, invoicing, rate management
Compliance Officer	Audit logs, retention, access review
Firm Admin	Users, roles, connectors, AI settings
External Counsel	Limited matter access
Client Viewer	Read-only matter visibility (optional)

2. Product Goals

2.1 Must Have (Phase 1 — UI Shell)

- Desktop-style application shell (Cursor-inspired)
- Resizable panes with persistent layout
- Monochrome light mode (primary); dark mode scaffold
- All 13 primary navigation modules with mock data
- Matter workspace with multi-pane layout
- Mock Legal AI chat with citation panel
- Mock document, email, workflow, billing, admin screens

2.2 Must Have (Production)

- Source-backed AI answers with verification status
- Citation engine with approval workflow
- Matter-level and document-level permissions
- Complete audit trail
- Human approval before email send
- South African jurisdiction awareness
- Private deployment packaging

2.3 Must Not

- Silently invent legal authorities
- Send emails without human approval
- Leak data across firm boundaries
- Present unverified AI claims as verified

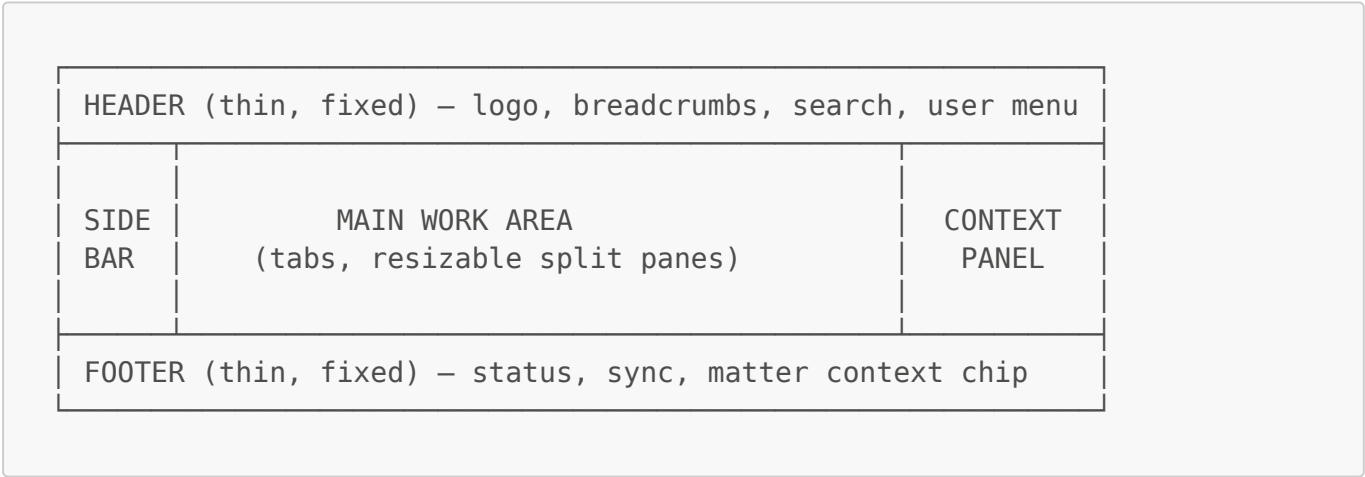
3. Primary Navigation Modules

#	Module	Description
1	Dashboard	Firm activity, deadlines, approvals queue, recent matters
2	Matters	Matter list, filters, create/open matter

#	Module	Description
3	Legal AI	Matter-aware AI chat with citations
4	Research	Legislation, case law, internal corpus search
5	Documents	Upload, review, versioning, folders
6	Email Discovery	Mailbox connect, import, thread review, draft replies
7	Workflows	Approval queues, status tracking, escalation
8	Time & Billing	Time entries, rates, invoices, matter summaries
9	Calendar	Deadlines, hearings, meetings
10	Tasks	Matter-linked tasks, assignments, due dates
11	Templates	Legal document templates
12	Admin	Firm configuration, users, AI, connectors
13	Audit Logs	Immutable action history

4. Application Shell

4.1 Layout Regions



4.2 Shell Requirements

- Fixed thin header and footer (see [UI-UX-SPEC.md](#))
- Collapsible left sidebar (icon-only collapsed state)
- Resizable right context panel
- Resizable main work area with horizontal and vertical splits
- Tabbed workspace within main area
- Command palette (**Cmd/Ctrl+K**)
- Keyboard shortcuts for navigation and pane focus
- Persistent user layout preferences (localStorage → DB in Phase 2)

5. Dashboard

Purpose: At-a-glance firm activity.

Contents (mock → live):

- Matters requiring attention
- Pending approvals (documents, emails, opinions, time)
- Upcoming deadlines (7 / 30 days)
- Recent AI sessions
- Unbilled time summary
- Quick actions: New matter, New time entry, Open Legal AI

6. Matter Workspace

The Matter Workspace is the **heart of the platform**.

6.1 Matter Data Model (functional)

Each matter includes:

- Matter summary (reference, title, status, practice area)
- Parties (clients, opponents, third parties)
- Assigned lawyers and roles
- Documents (folder tree, versions)
- Emails (threads linked to matter)
- Notes
- Tasks and deadlines
- Time entries
- Billing records
- AI chats linked to matter
- Research history
- Citation packs
- Approval history
- Audit trail

6.2 Multi-Pane Layout

Pane	Position	Resizable	Contents
Matter nav	Left	Yes	Folders, emails, tasks, timeline
Main content	Centre	Yes	Document, email, chat, research
Context	Right	Yes	AI assistant, citations, facts, risks, approvals

All three panes must resize independently. Minimum widths enforced to prevent collapse.

7. Legal AI Engine

7.1 Behaviour Rules

The Legal AI is a **legal work assistant**, not a general chatbot.

Every answer must support:

Field	Requirement
Source references	Linked to corpus chunks
Confidence level	Low / Medium / High
Citation status	Verified / Unverified / Not found
Paragraph references	Where applicable
Verification label	Visible in UI
Incomplete material warning	When corpus is insufficient
Matter context	Scoped to active matter when set
Jurisdiction	South Africa default; configurable
Human review	Required before external use

7.2 Hard Rule — No Invented Authorities

If no source is found in the available corpus, the UI must display:

Source not found in the available corpus.

The AI must not silently fabricate case names, statute sections, or citations.

7.3 Technical Direction (later phases)

- LangGraph for stateful agent orchestration
- Ollama for local models (private deployment)
- Qdrant for vector retrieval
- OpenSearch for full-text legal search
- Optional LiteLLM for external model routing

8. Citation Engine

Dedicated citation subsystem — not an afterthought.

8.1 Capabilities

- Extract citations from judgments, legislation, uploaded documents
- Identify legislation references, case references, paragraph/page refs
- Link AI claims to source chunks
- Show citation confidence
- Flag unsupported claims
- Lawyer actions: Approve / Reject / Re-check
- Generate citation pack per matter

8.2 Citation Panel (beside AI answers)

Each citation displays:

Element	Description
Source title	Document or authority name
Document type	Judgment, statute, contract, email, etc.
Reference	Paragraph, section, or page
Excerpt	Quoted source text
Confidence	Score 0–100
Verification status	Verified / Pending / Rejected
Open source	Button to open full document at reference

9. Email Discovery

First-class module — not a bolt-on.

9.1 MVP Scope

- Architecture and UI complete
- Connectors mocked (Gmail, Microsoft 365 designed for later)

9.2 Features

- Connect mailbox
- Import selected folders
- Link emails to matters
- Extract attachments into document store
- Identify parties
- Correspondence timeline
- Deadline and undertaking extraction
- Thread summarisation
- Draft replies (AI-assisted)
- **Human approval required before send**
- Audit every email action

9.3 Hard Rule

AI may draft emails. A human **must approve** before sending.

10. Workflow and Approvals

10.1 Workflow Types

- Document approval
- Email approval
- Legal opinion approval
- Contract review approval
- Billing approval
- Task assignment and escalation

10.2 Status Model

Draft → In Review → Changes Requested → Approved → Sent / Filed → Archived

10.3 Workflow Record

- Reviewer comments
- Status history (immutable append)
- Due dates
- Escalation rules
- Linked entity (document, email, opinion, time entry)

11. Time Management and Billing

11.1 Rate Types

- User hourly rates
- Role rates
- Matter-specific rates

11.2 Time Entry Types

- Manual entry
- Timer-based entry
- AI-suggested entry (requires approval)

11.3 Linkage

Time entries link to: matter, client, task, document, email, AI session.

11.4 Billing

- Billable vs non-billable
- Approval of time entries
- Invoice preparation
- Matter billing summary
- Export to accounting (future)

12. Research Module

- Full-text search (OpenSearch) across legislation, judgments, firm corpus
- Vector search (Qdrant) for semantic retrieval
- Research history per matter
- Save results to matter
- Export to citation pack

Jurisdiction default: South Africa (Constitution, Acts, regulations, SA case law).

13. Admin Console

Section	Configuration
Users	CRUD, invite, deactivate
Roles & permissions	RBAC matrix
Matters	Bulk admin, archive
Billing rates	User, role, matter overrides
AI model settings	Ollama models, timeouts, concurrency
Email connectors	OAuth credentials, sync schedule
Document retention	Policies per matter type
Audit logs	Search, export
Workflow templates	Reusable approval flows
Firm branding	Logo, name (private deployment)
Backup settings	Schedule, destination
Deployment health	Service status, disk, model availability

14. Non-Functional Requirements

Category	Requirement
Performance	Pane resize < 16ms; list virtualisation for 10k+ documents
Accessibility	Keyboard navigation; focus rings; ARIA on interactive controls
Responsiveness	Desktop-first; minimum 1280×720
Resizability	All primary panes independently resizable
Persistence	Layout preferences survive refresh
Security	See SECURITY.md
Audit	All sensitive actions logged immutably
Localisation	English (SA) first; Afrikaans later

15. Phase 1 Acceptance Criteria

Phase 1 is complete when:

- ☐ Next.js app runs locally with App Router + TypeScript strict
- ☐ Cursor-style shell: thin header, thin footer, pill controls, 1px borders
- ☐ Monochrome light mode implemented per UI spec
- ☐ Left sidebar, resizable main area, resizable right panel
- ☐ All 13 nav modules reachable with mock data
- ☐ Matter workspace with three resizable panes
- ☐ Mock Legal AI chat with citation panel
- ☐ Mock email discovery, workflow, billing, admin screens
- ☐ Command palette functional (navigation only)
- ☐ Layout preferences persist in localStorage
- ☐ No backend required beyond static mock JSON

16. Out of Scope (Phase 1)

- Real authentication (Better Auth — Phase 2)
- Database schema and migrations (Phase 2)
- Document ingestion (Phase 3)
- RAG / search (Phase 4)
- Citation verification engine (Phase 5)
- Email connectors (Phase 6)
- Production deployment (Phase 9)

17. Success Metrics (Post-Launch)

Metric	Target
Time to open matter workspace	< 2s
AI answer with citations	100% show verification status
Unsupported claims flagged	100% when no source
Email send without approval	0 (blocked by system)
Cross-firm data leakage	0
Lawyer NPS (pilot firms)	≥ 40